IST 691: Research Methodology and Scientific Writing Skills

Ethics in Research

Denis L. Nkweteyim

Outline

- Introduction to Ethics
- Ethics in Research

Introduction

- Ethics
  - Also known as moral philosophy, ethics is a branch of philosophy that involves systematizing, defending, and recommending concepts of right and wrong behaviour

Introduction

- One view of ethics
  - Rules for distinguishing between right and wrong
  - Examples
    - The Golden Rule ("Do unto others as you would have them do unto you")
    - A code of professional conduct like the Hippocratic Oath ("First of all, do no harm")
    - A religious creed like the Ten Commandments ("Thou Shalt not kill…")
  - One problem with this view
    - People may recognize some common ethical norms but interpret, apply, and balance them in different ways
    - For example, two people could agree that murder is wrong but disagree about the morality of abortion because they have different understandings of what it means to be a human being

Introduction

- A second view of ethics
  - Focuses on the disciplines that study standards of conduct
    - Philosophy, theology, law, psychology, sociology, etc.
  - Example
    - Medical ethicist studies ethical standards in medicine
- Yet another view of ethics
  - Ethics is a method, procedure, or perspective for deciding how to act and for analyzing complex problems and issues
  - E.g., the complex issue of global warming
    - An economist might examine the cost and benefits of various policies related to global warming
    - An environmental ethicist could examine the ethical values and principles at stake
    - Etc.

Introduction

- Different disciplines, institutions, and professions have standards for behaviour that suit their particular aims and goals
  - These standards also help members of the discipline to coordinate their actions or activities and to establish the public's trust of the discipline
    - In the practice of medicine, law, engineering, and business, for example
  - Ethical standards also serve the aims or goals of research and apply to people who conduct scientific research or other scholarly or creative activities
    - Research ethics is the discipline that studies these norms

Outline

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Introduction to Ethics in Research

- What are research ethics?
  - The moral principles and codes of conduct that guide how studies are designed, executed, and reported to ensure integrity and the protection of participants
- Objectives
  - Protect human and animal participants from harm
  - Ensure data reliability and scientific integrity
  - Maintain public trust in scientific progress

Why worry about ethics in research?

- I'm a good person. Why do I need to worry about research ethics?
  - True, we can't do much about the bad person who is determined to do evil things
  - However, research ethics isn't just – or even mostly - about bad people doing bad things
  - Research ethics is also about imperfect people doing imperfect things for various reasons

Why worry about ethics in research?

- Research ethics is also about imperfect people doing imperfect things for various reasons
  - Socially acceptable practice we later deem wrong
  - Accident
  - Misdemeanour – i.e., minor wrongdoing that we try to justify (e.g., taking shortcuts)
  - Missing something, especially with new methods or technology
  - One can become involved in research ethics violations through the wrongs of others
  - Sometimes the right thing to do just isn't clear
  - Professional pressures
    - Publish or perish; Tenure/retaining a job; "Keeping up" with peers; Securing grants; Being first to a discovery, etc.
  - Others

Why worry about ethics in research?

- Doing good for humans, animals, the planet, future generations, etc. via the pursuit of truth and knowledge
  - Our duty to respect individuals
  - Our possible duties to animals
  - Our obligations to society
- Various obligations derive from these foundations, and they can be in tension with one another

Why it is important to adhere to ethical norms in research

1) Norms promote the aims of research, such as knowledge, truth, and avoidance of error
   - ✓ For example, prohibitions against fabricating, falsifying, or misrepresenting research data promote the truth and minimize error
2) Ethical standards promote the values that are essential to collaborative work, such as trust, accountability, mutual respect, and fairness
   - ✓ Since research often involves much cooperation and coordination among many different people in different disciplines and institutions
   - ✓ Example
     - Guidelines for authorship, copyright and patenting policies, data sharing policies, and confidentiality rules are designed to protect intellectual property interests while encouraging collaboration
     - Most researchers want to receive credit for their contributions and do not want to have their ideas stolen or disclosed prematurely

Why it is important to adhere to ethical norms in research

3) Many of the ethical norms help to ensure that researchers can be held accountable to funders of the research
4) Ethical norms in research also help to build public support for research, e.g., willingness of people to fund research projects if they can trust the quality and integrity of research
5) Many of the norms of research promote a variety of other important moral and social values
   - ✓ Social responsibility, human rights, animal welfare, compliance with the law, and public health and safety
   - ✓ Ethical lapses in research can cause significant harm
     - Researcher who fabricates data in a clinical trial may harm or even kill patients
     - Researcher who fails to abide by regulations and guidelines relating to radiation or biological safety may jeopardize his health and safety or the health and safety of staff and students

Foundational Principles

- Respect for Persons (Autonomy)
  - Acknowledging individuals as independent agents and protecting those with diminished autonomy
    - Voluntary Participation: Subjects must be free to join or leave a study at any time without penalty
    - Informed Consent: Participants must understand the purpose, risks, and benefits before agreeing to join
    - Protection of Vulnerable Groups: Extra safeguards for children, those with diminished capacity, or those in compromised positions.
- Beneficence & Non-Maleficence
  - Do No Harm: Minimizing physical, psychological, social, or legal risks.
  - Maximizing Benefit: Research must provide value that outweighs its risks.
- Justice
  - Fair distribution of the burdens and benefits of research
    - Avoid exploiting "convenient" populations (e.g., marginalized groups)
    - Ensure results benefit the communities involved

Other Core Issues

- Integrity and Honesty in Scientific Practice
  - Researchers must truthfully report data and results without fabrication, falsification, or plagiarism
  - Avoiding Research Misconduct
    - Fabrication: Inventing data or results
    - Falsification: Manipulating research materials or omitting findings to fit a hypothesis
    - Plagiarism: Appropriating others' work or ideas without credit
    - Responsible Publication
      - Advancing scholarship rather than just a career
      - Avoiding duplicative publications

Other Core Issues – Example

- Ethical guidelines stemming from research as the pursuit of truth
  - “The truth, the whole truth, and nothing but the truth.”
    - The truth: Be honest about your research
    - The whole truth: Omission of parts of research findings might constitute research misconduct or violate other moral norms
    - Nothing but the truth: It’s also dishonest to puff up one’s results by adding irrelevant or misleading information, or overstating their significance

Other Core Issues

- Privacy, Confidentiality and Anonymity
  - Protecting the privacy of participants
    - Anonymity: Data is collected without any identifying information; Even the researcher cannot identify the subject
    - Confidentiality: The researcher knows the identity but keeps it private from public disclosure.
    - Data Protection: Secure storage of digital files (often using encryption)

Ethical dilemmas

- Some fields of research (e.g., psychology, medicine) create a special set of dilemmas
  - On the one hand, the researcher feels an obligation to conduct research
  - On the other hand, such research may subject research participants to stress, pain, aggression, deception
- Ethical dilemma
  - The researcher's conflict in weighing the potential cost to the research participants against the potential gain to be accrued from the research project
  - In weighing the pros and cons of such a question, researcher must give primary consideration to the welfare of the participant
  - Unfortunately, such judgement can only be subjective, and so should not be made solely by the researcher and his/her colleagues

Institutional Oversight

- Institutional Review Board (IRB)
  - Most human-subject research requires prior approval from an IRB, also known as research ethics committee
  - Provides a core protection for human research participants through advance and periodic independent review of the ethical acceptability of proposals for human research
  - Charged with the responsibility of reviewing, prior to its initiation, all research involving human participants
  - Concerned with protecting the welfare, rights, and privacy of human subjects
  - Has the authority to approve, exempt, disapprove, monitor, and require modifications in all research activities that fall within its jurisdiction
  - Continuing Review: IRB monitors ongoing studies for "mission creep" (going beyond stated objectives).
  - Conflict of Interest (COI): Researchers must disclose financial or personal interests that could influence findings.

Codes and Policies for Research Ethics

- Given the importance of ethics in research, many professional associations, government agencies, and universities have adopted specific codes, rules, and policies relating to research ethics